

Class Notes – Computer Graphics

Date: 06/08/2026

1. Computer Graphics vs Image Processing

Computer Graphics: Creating and generating images or visual content using computers.

Image Processing: Manipulating or improving an existing image.

Example: Creating a 3D model → Computer Graphics; enhancing a photograph → Image Processing.

2. Geometric Primitives

Basic shapes used to create graphics:

Point, Line, Circle, Rectangle, Polygon, Ellipse, Curves

Complex objects can be created by combining these basic primitives.

3. Graphical Frameworks in Java

Java provides several frameworks for developing graphical applications:

AWT (Abstract Window Toolkit): Basic GUI components and graphics.

Swing: Provides advanced and platform-independent GUI components.

JavaFX: Modern framework for creating rich graphical applications.

4. Design Pattern for User Interfaces

A design pattern is a reusable approach to solving a common software-design problem.

MVC (Model–View–Controller) is commonly used for user interfaces:

Model: Manages data and logic.

View: Displays the user interface.

Controller: Handles user input and connects the Model and View.

5. Static vs Interactive Graphics

Static Graphics: The image remains unchanged and does not respond to user actions.

Interactive Graphics: The user can interact with and change the graphics through actions such as clicking, dragging, or typing.

6. Curves and Calculus

Calculus helps us understand and represent curves mathematically.

Differentiation helps find the slope and rate of change of a curve.

Integration helps calculate areas and other quantities related to curves.

Curves in computer graphics can be represented using mathematical equations.

7. SSH and HTTP for Accessing GitHub

SSH (Secure Shell):

Provides secure authentication and communication with GitHub.

Uses SSH keys instead of entering a password repeatedly.

Example:

```
git@github.com:username/repository.git
```

HTTP/HTTPS:

Uses a web-based protocol to communicate with GitHub.

Repository URLs generally start with `https://github.com/`.

8. Tools Used for SSH Operations

Common SSH-related tools/commands:

`ssh` – connect/authenticate with a remote server.

`ssh-keygen` – generate SSH key pairs.

`ssh-agent` – manages private keys.

`ssh-add` – adds a private key to the SSH agent.

`ssh -T git@github.com` – tests GitHub SSH authentication.

9. Public Key and Private Key

SSH uses a key pair:

Public Key: Can be shared and added to GitHub.

Private Key: Must be kept secret on your computer.

The two keys work together to securely verify your identity.